

Isaac Mubiru

isaacmubiru99@gmail.com | linkedin.com/in/isaac-mubiru-3bb728174 | github.com/zac-09 | Kampala, Uganda

PROFESSIONAL SUMMARY

Senior Backend Engineer with 6+ years building scalable backend systems, APIs, and cloud infrastructure using Node.js and TypeScript. Led zero-downtime migration of 20+ production applications from Parse/MongoDB to Firebase/GCP on AWS, cutting infrastructure costs by \$5,000/month. Experienced in designing real-time data pipelines, microservices architecture with Kubernetes, and automation infrastructure. Strong track record of remote collaboration with US companies and cross-functional teams.

CORE COMPETENCIES

Node.js / TypeScript

AWS Infrastructure

Kubernetes / Docker

MongoDB / NoSQL

Backend APIs

Real-time Data Pipelines

Microservices Architecture

Zero-Downtime Migrations

WORK EXPERIENCE

MTailor

Software Engineer (US Remote)

- Led end-to-end migration of 20+ production applications from Parse/MongoDB to Firebase/GCP on AWS infrastructure, ensuring zero downtime while serving live traffic
- Designed and implemented real-time two-way sync pipeline between MongoDB and Firestore using Node.js and Google Pub/Sub for message processing
- Built backend automation for 3D visualization feature using video overlay and ffmpeg, increasing buyer conversion
- Implemented Express Shipping integration, generating +\$40 revenue per order
- Reduced infrastructure costs by \$5,000/month by migrating all services off AWS to optimized GCP architecture

Sep 2022 – Present

MTailor

Contractor (US Remote)

- Built scalable migration scripts to read data from MongoDB and write to Firestore with complex processing logic
- Authored technical documentation for Firebase SDKs (Node.js, Python) and trained Ops team on new infrastructure

Mar 2022 – Jul 2022

Mind2matter.co

Software Engineer (US Remote)

- Built backend APIs and infrastructure for DeFi applications using Web3 and Node.js
- Delivered React UIs from Figma designs under tight deadlines, reporting directly to CTO

Jan 2022 – Jul 2022

Dr Wealth

Software Engineer (Singapore Remote)

- Optimized PWA delivering real-time stock market prices, building backend APIs with Express.js and Firebase Cloud Functions
- Built automation jobs to query 2M+ Firestore records, keeping customer prices current from Morningstar APIs

Aug 2021 – Jan 2022

CodeBits

Software Engineer (Uganda Remote)

- Led team of 4 developers building legal tech systems for NGOs in Uganda
- Architected microservices backend using Apache Kafka, Docker, and Kubernetes for FIDA Uganda case management
- Built cross-platform mobile app using React Native/Expo with Firebase push notifications (iOS & Android)
- Developed USSD service communicating with Node.js backend via gRPC

Jan 2020 – Jul 2021

EDUCATION

Bachelor of Science in Computer Engineering — Makerere University
Kampala, Uganda 2018 – 2022

SKILLS

Languages: TypeScript, JavaScript, Python, Go, Rust **Backend:** Node.js, Express.js, Firebase Cloud Functions, REST APIs
Cloud: AWS (S3, EC2, EBS), GCP, Firebase, Heroku **Infrastructure:** Kubernetes, Docker, Apache Kafka, gRPC
Databases: MongoDB, Firestore, SQL, Redis **Frontend:** React, React Native, Tailwind CSS, HTML/CSS **Tools:** Git, Linux, GitHub Actions